

INTRODUCTION

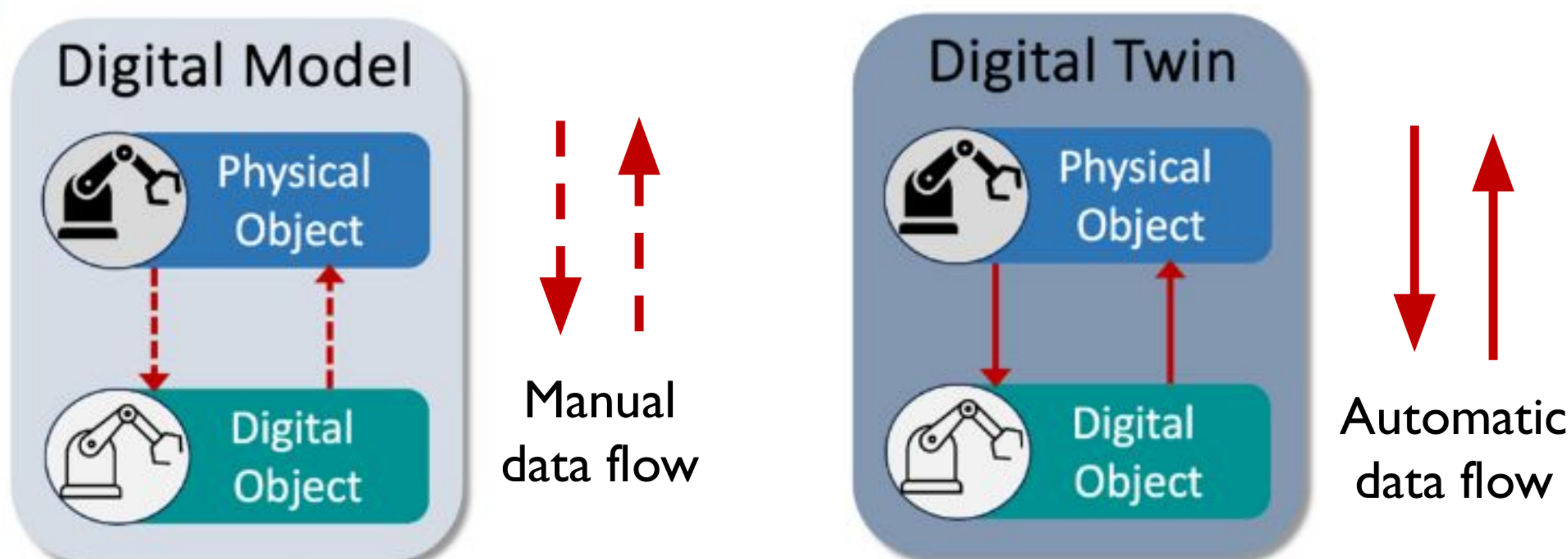
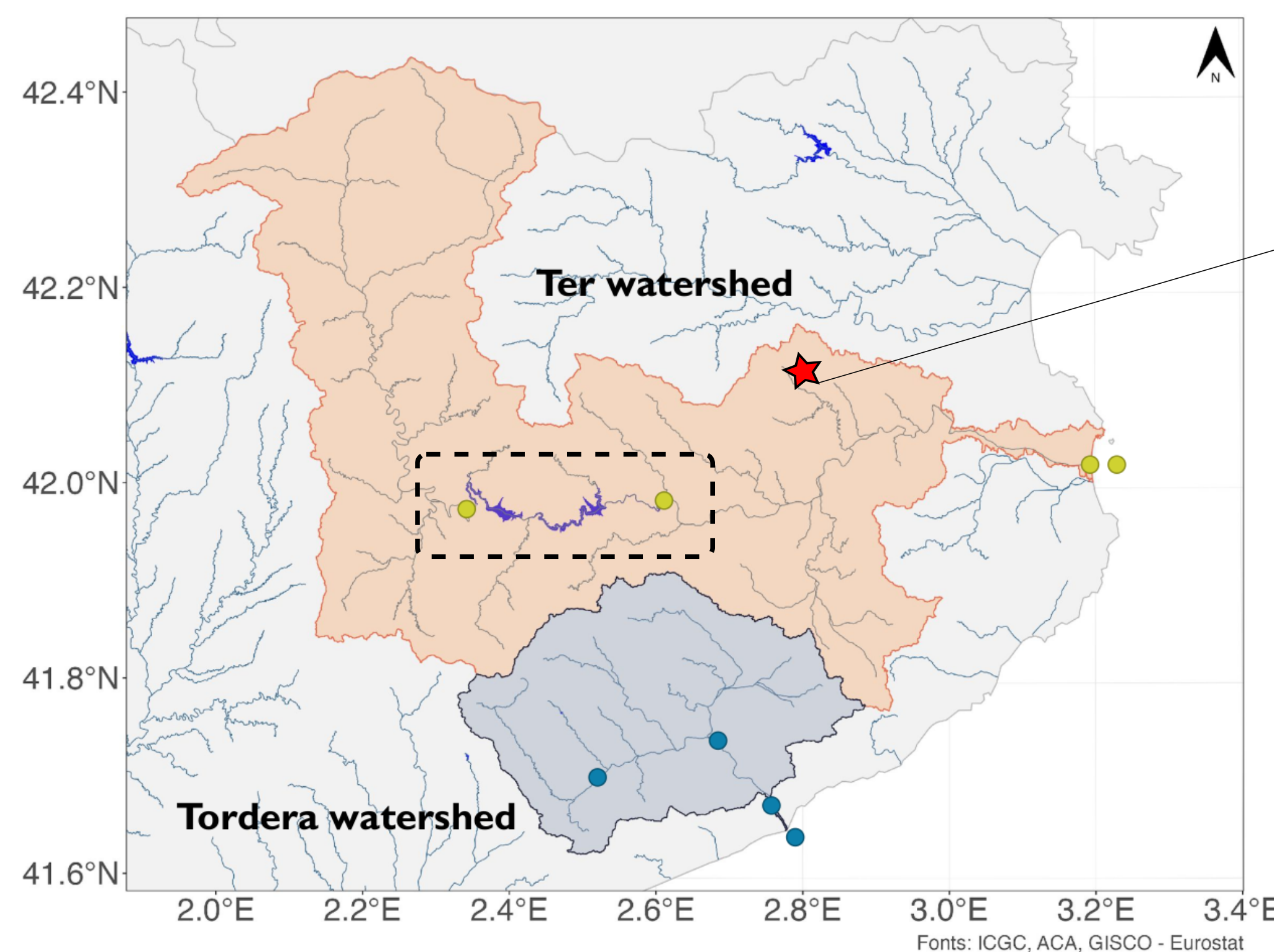


Figure 1. Digital Twin vs Digital Model

- Co-development with water managers
- Prediction of water quantity and quality
- Two comparable watersheds
- Includes reservoirs and coastal areas.



Next
GLEON
meeting
will be here!

Main source
of drinking
water for
Barcelona
and Costa
Brava

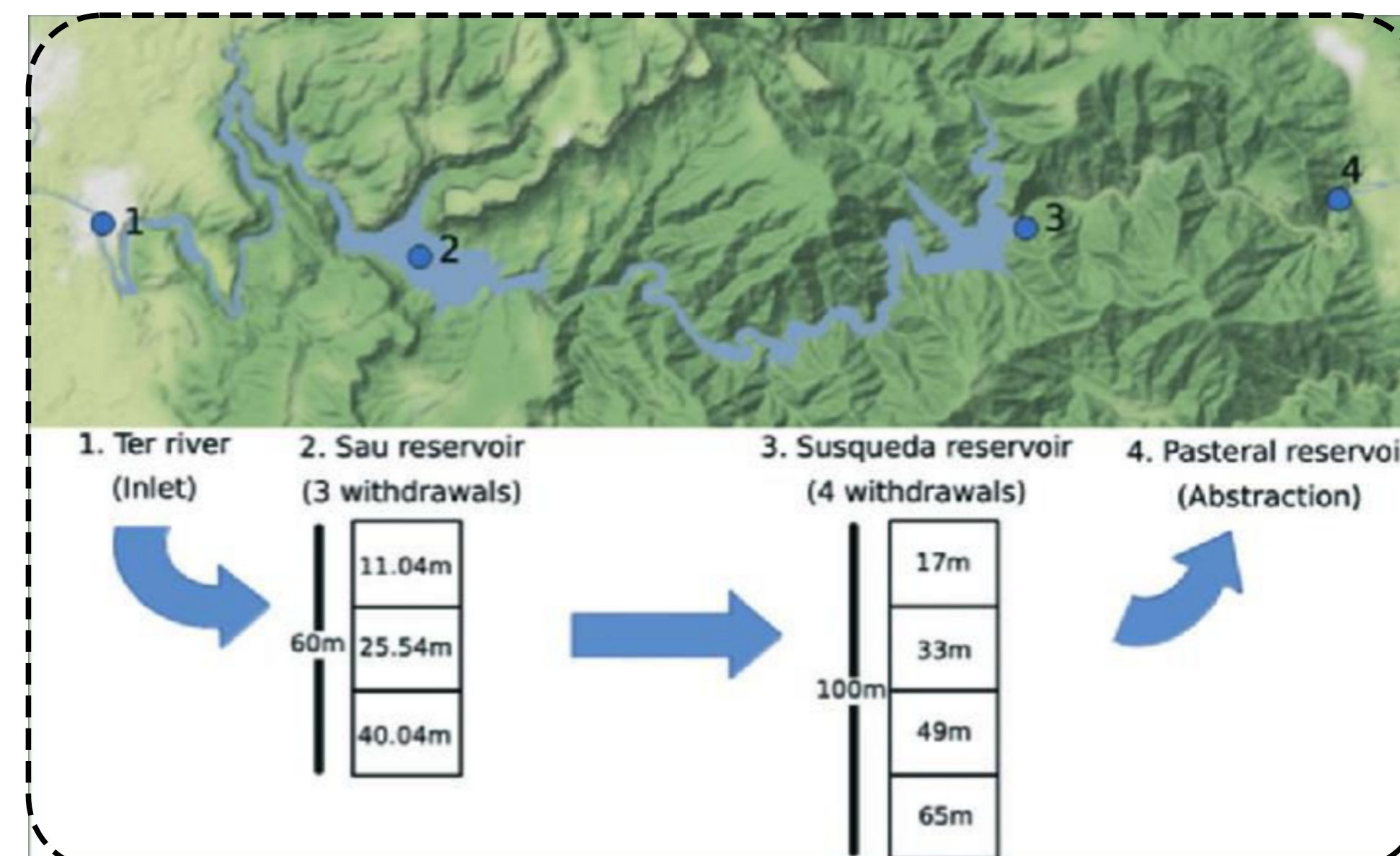
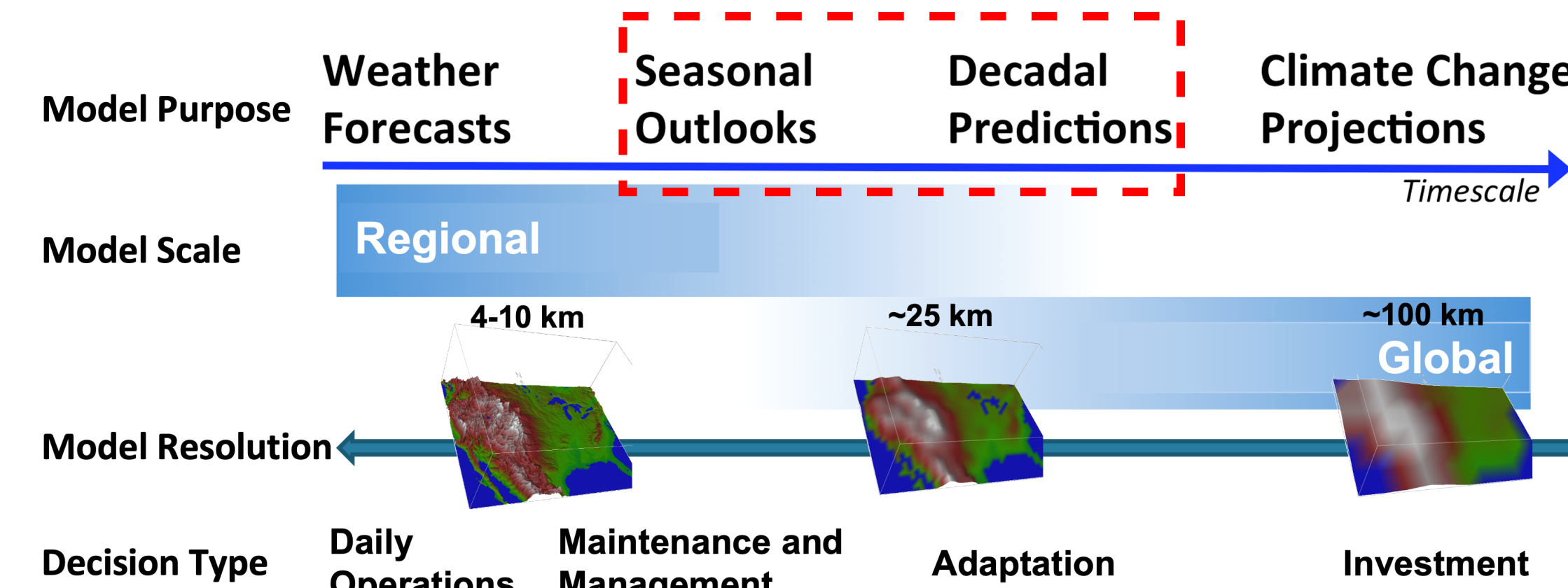


Figure 2. Study sites, the Ter and Tordera watersheds

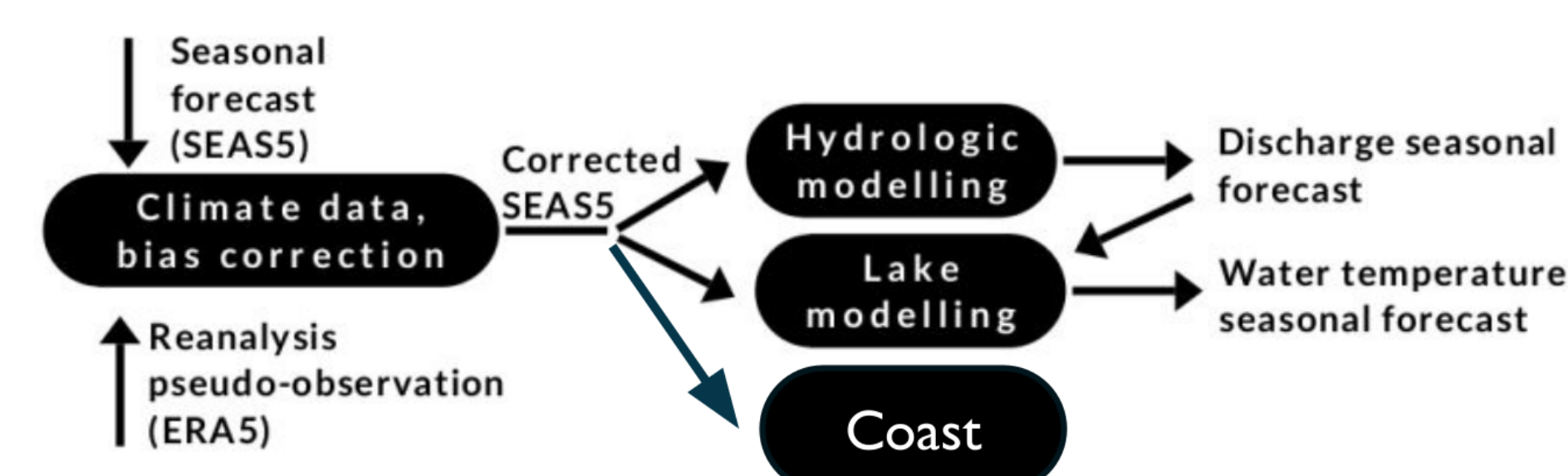
METHODS



Adapted from: Tye MR et al. (2025) HESS

Figure 3. Temporal scales for water management decisions

- There are multiple temporal scales
- Which ones are more relevant for strategic planning and policy making?
- Seasonal and decadal prediction
- Drought and humid seasons



Mercado-Bettín et al. (2021) Water Research

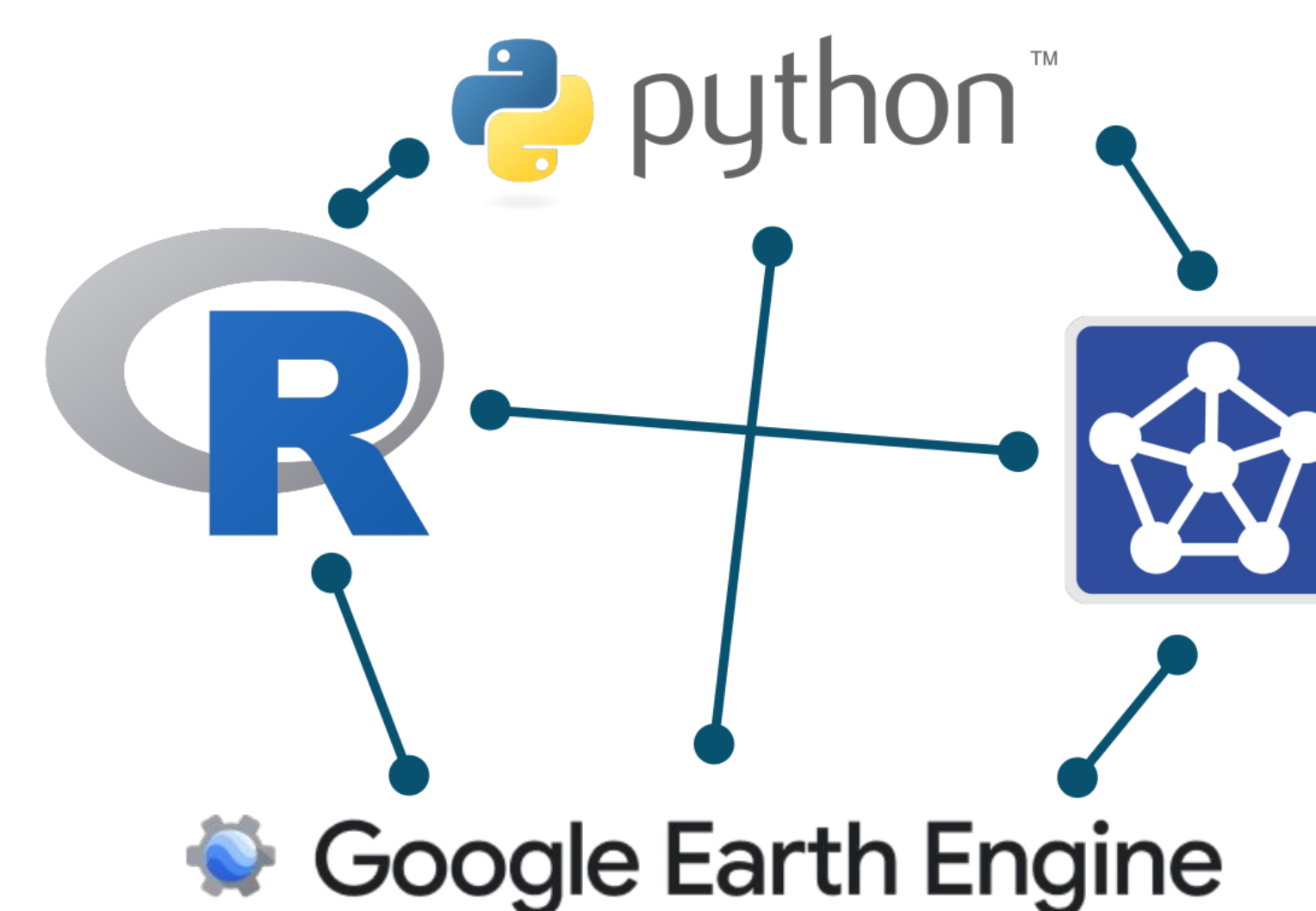


Figure 4. Workflow and tools to be used in the Digital Twins

- Water balance (streamflow and volumes) and water quality (temp., nitrogen, sediments)
- River modeling (R), lake (GLM and R), and coastal modeling (DELFL).
- Knowledge based machine learning.

RESULTS

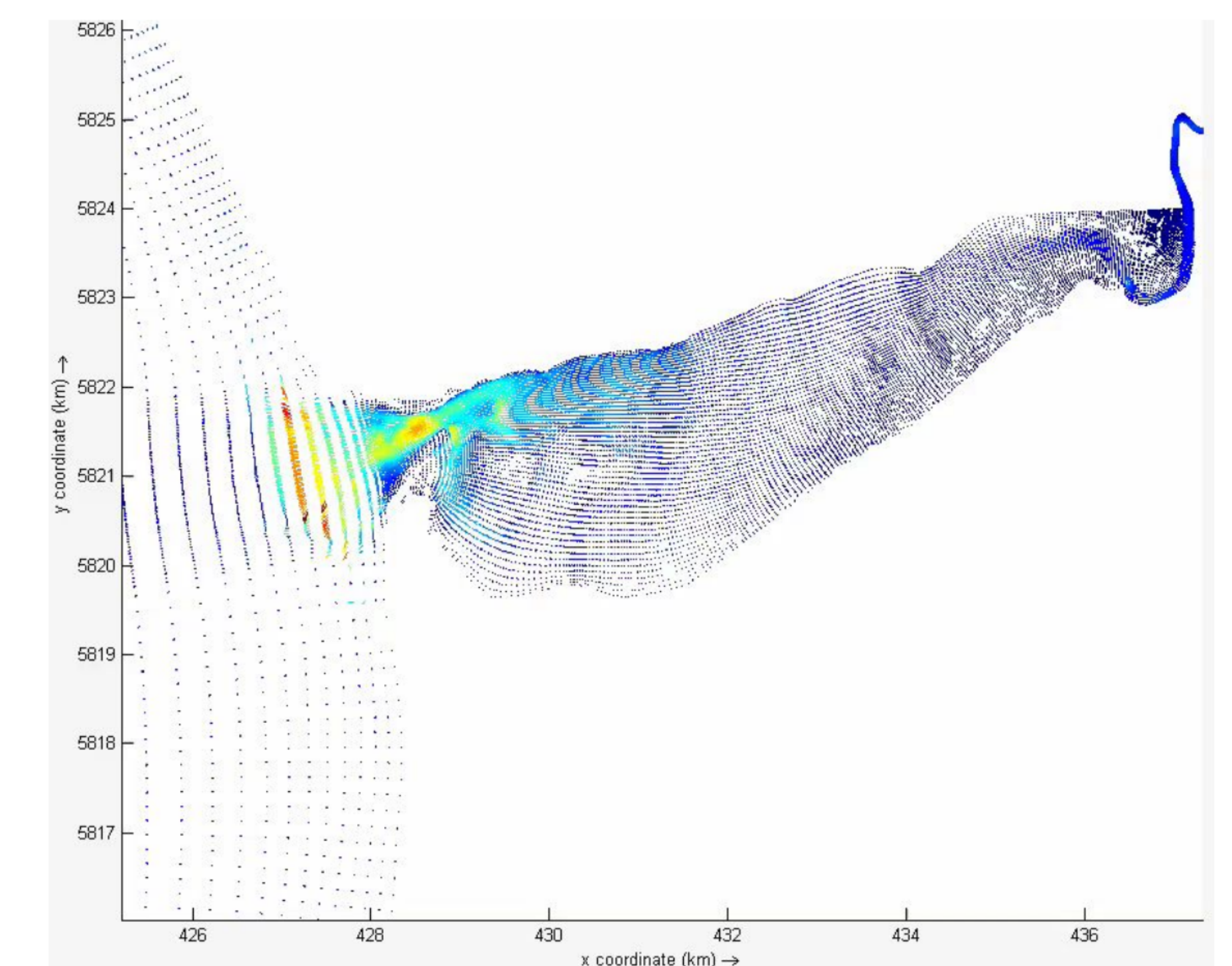
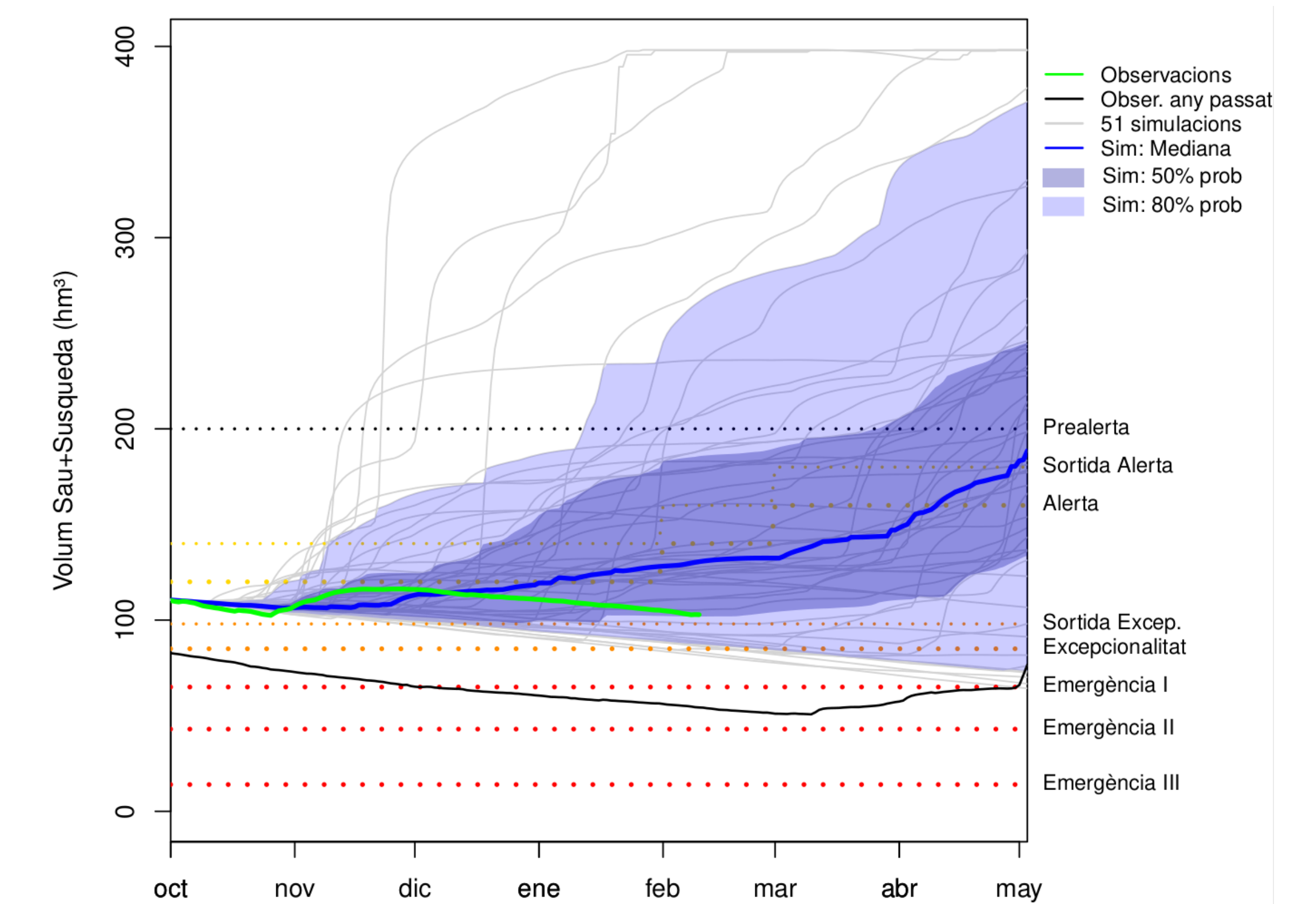


Figure 5. Example of seasonal prediction of water balances in reservoirs and coastal sedimentation, to support water management decisions.

Tool deployment (under development):

https://nereidaca.github.io/seasonal_prediction/

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Water Agency